ANDREW IMOUKHUEDE Team 308 Power Budget

Team Number:	308						
Project Name:							
Team Member Names:	**Andrew, Sam B, Sam M, Jacob D, Adrian P, Mo**						
Version:	**1.0**						
A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	ESP-32	ESP32-S3-WR	+3.0V - 3.6V	1	1000	1000	mA
	IMU	BNO055	+2.4V - 3.6V	1	20	20	mA
					1020	**1020**	**mA**
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+3.3V Power Rail	**Component Name**	**Part Number**	**SupplyVoltageRange**	**#**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
	ESP-32	ESP32-S3-WR	+3.0V - 3.6V	1	1000	1000	mA
	IMU	BNO055	+2.4V - 3.6V	1	20	20	mA
						0	mA
						0	mA
					Subtotal	1020	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	1275	mA
c4. Regulator or Source Ch	3.3V Regulator	AP63203WU-7	+3.8V - 36V	1	2000	2000	mA
					Total Remaining Current Available on 3.3V Rail	725	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously.							
External Power Source 1	**Component Name**	**Part Number**	**SupplyVoltageRange**	**Outpu**	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	full part numbe	110VAC	+24V	5000	5000	mA
Power Rails Connected to	3.3V Regulator	A4402KLPTR-T	1.8V -3.3V	1	2000	2000	mA
					Total Remaining Current Available on External Power Source 1	3000	mA